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LIST OF ABBREVIATIONS

Abbreviation	Full Term	Meaning / Explanation
AI	Artificial Intelligence	Computer systems designed to perform tasks commonly associated with human intelligence
API	Application Programming Interface	A defined interface through which software components communicate
CV	Curriculum Vitae	A document describing a candidate's qualifications and experience
E2E	End-to-End	Testing of a complete workflow across integrated components
HITL	Human-in-the-Loop	Automation in which humans retain oversight or approval authority
JD	Job Description	A description of a job's responsibilities and requirements
JWT	JSON Web Token	A compact token format used for authentication and authorization
MRR	Mean Reciprocal Rank	A ranking metric based on the position of the first relevant item
nDCG	Normalized Discounted Cumulative Gain	A ranking metric that accounts for relevance and position
NLP	Natural Language Processing	Methods for processing and analyzing human language
OCR	Optical Character Recognition	Extraction of machine-readable text from images or scanned documents
RBAC	Role-Based Access Control	Authorization based on assigned user roles
REST	Representational State Transfer	An architectural style for networked APIs
TF-IDF	Term Frequency–Inverse Document Frequency	A weighting method for representing text terms
UAT	User Acceptance Testing	Validation of whether the system satisfies intended user workflows

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

IT recruitment requires candidates and recruiters to compare many details, including technical skills, experience, language, location, and working conditions. Candidates must identify suitable vacancies, while recruiters must review CVs and decide which applicants deserve closer attention. As the number of jobs and profiles grows, doing this manually becomes slow and inconsistent [10].

Most job portals support posting, searching, and applying, but they do not fully explain why a job fits a candidate or why one applicant ranks above another. CareerFit therefore treats CV–JD matching and personalized recommendation as related but separate tasks. The first compares a specific CV and job; the second prioritizes jobs from a candidate's broader profile and preferences.

Automation can reduce repetitive work, but a similarity score should not directly trigger a consequential action. Consent, current application state, previous interactions, thresholds, quotas, and exceptional conditions also matter. CareerFit uses a Human-in-the-Loop design so that users can review important actions and administrators can trace what happened.

This thesis develops CareerFit IT AutoPilot, an IT-focused web platform that combines job discovery, CV and JD processing, matching, recommendation, Rocchio feedback, AutoFit policies, email actions, and audit records. The aim is not to replace recruitment judgment, but to provide understandable evidence and controlled assistance throughout the workflow.

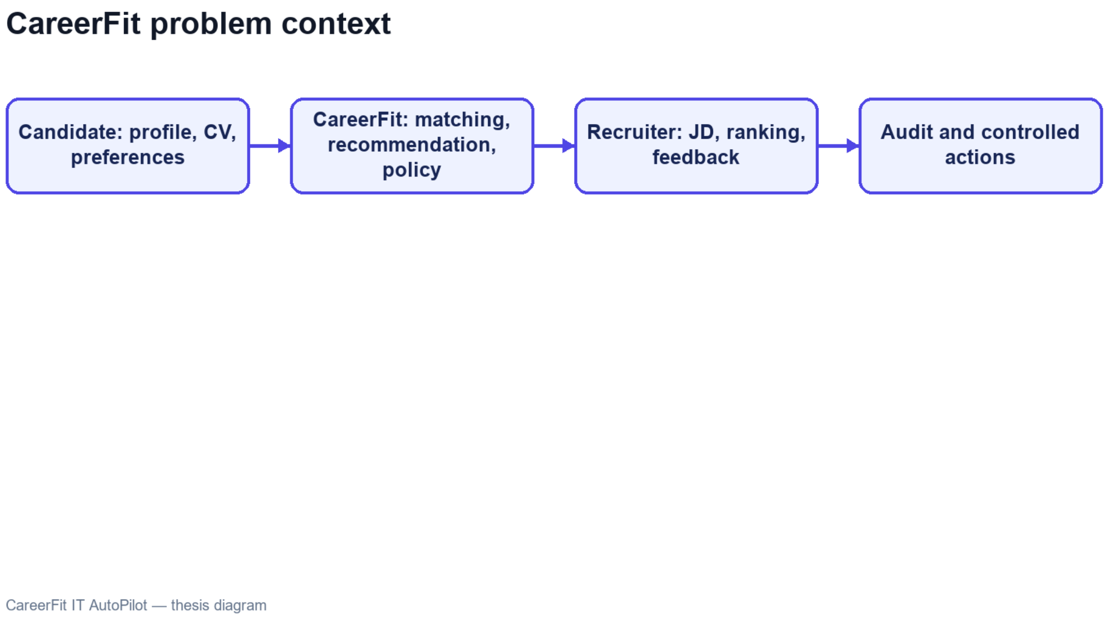

Figure 1.1. CareerFit problem context and principal information flows

1.2 Motivation

The first motivation is to reduce fragmentation. Candidates often move between job search, CV management, application history, and email, while recruiters switch between vacancy management, applicant review, candidate discovery, and reporting. CareerFit brings these activities into one role-based system and uses email as an additional action channel.

The second motivation is explainability. TF-IDF and cosine similarity do not capture meaning as deeply as modern embedding models, but their terms and weights can be inspected. This makes them suitable for an academic baseline in which the scoring process, limitations, and effect of feedback must be demonstrated clearly.

The third motivation is controlled adaptation and automation. AutoFit places policy checks between a score and an action, considering consent, thresholds, previous interactions, cooldown, quota, time zone, and quiet hours. Rocchio feedback then provides a transparent way to adjust later rankings from explicit positive and negative judgments without overwriting the original representation.

1.3 Objectives

1.3.1 Overall Objective

The overall objective of this thesis is to design, implement, and evaluate a web-based Human-in-the-Loop recruitment automation platform for the IT domain. The platform is intended to support CV–job description evaluation, personalized job recommendation, feedback-based adaptation, and policy-driven actions while preserving user control, explainability, security, and auditability.

1.3.2 Specific Objectives

To achieve the overall objective, the thesis defines the following specific objectives:

- Design a role-based web platform for guests, candidates, recruiters, and administrators, with appropriate public and authenticated workflows.
- Provide candidate profile and CV management, including document upload, form-based entry, text extraction, OCR fallback, validation, and quality feedback.
- Implement separate CV–JD matching and candidate-profile recommendation workflows using a shared text-normalization, TF-IDF, and cosine-similarity pipeline.
- Present normalized scores, relevance labels, matching reasons, validation signals, and stable ranking behavior to support interpretation by candidates and recruiters.
- Integrate explicit user feedback and apply the Rocchio relevance-feedback method to update learned representations and recompute rankings reproducibly.
- Implement AutoFit policies that convert matching results into controlled outcomes according to consent, thresholds, interaction state, quota, cooldown, time-zone, and quiet-hour rules.
- Support expiring email-action links with hashed token storage, a non-mutating confirmation page, POST execution, action-state tracking, and audit logging.
- Evaluate the algorithmic behavior and system workflows using reproducible experiments, automated tests, scripted acceptance scenarios, and security and reliability checks appropriate to the project scope.

1.4 Scope of the Thesis

This thesis focuses on an academic prototype for recruitment in the information technology domain. Its functional scope includes a public job portal; authenticated candidate, recruiter, and administrator workspaces; CV and profile management; job and employer management; CV–JD matching; personalized job recommendation; application workflows; explicit relevance feedback; AutoFit policy configuration; notifications and actionable email; analytics; and audit records.

The data-processing scope includes extraction of text from supported CV documents, OCR fallback for image-based content where the configured runtime permits it, hard and soft validation, bilingual-oriented text normalization, TF-IDF vectorization, cosine-similarity scoring, score normalization, relevance labeling, and Rocchio-based feedback updates. PostgreSQL is used as the primary data store, and database evolution is managed through Flyway migrations. The implemented system follows a client–server architecture with a React and TypeScript frontend and a Spring Boot backend exposing REST APIs.

The thesis does not attempt to implement a complete enterprise applicant tracking system. Interview scheduling, offer management, payroll, and full employee lifecycle management are outside the core scope. The system does not automatically submit applications to external third-party job boards, and it does not use a large language model to make unrestricted recruitment decisions. Distributed microservices, large-scale message brokers, and semantic embedding models may be discussed as future extensions but must not be presented as implemented core components unless they are added and verified before the final report is submitted.

The evaluation scope is also bounded. Controlled or synthetic datasets may be used to demonstrate whether the Rocchio feedback mechanism changes rankings in the expected direction, but such results do not establish recruitment quality in a production population. Any scraped, seeded, or synthetic data used in experiments must be identified by provenance and purpose. Claims about usability, performance, security, or production readiness will be limited to the test environment, participants, procedures, and evidence actually available at the time of the final evaluation.

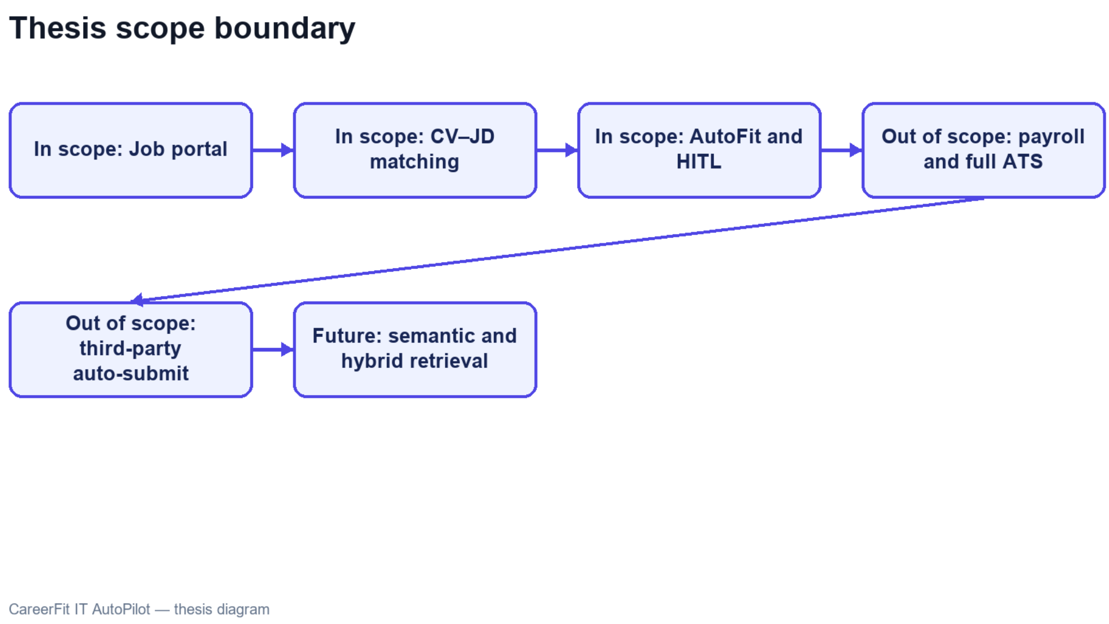

Figure 1.2. Scope boundary of the CareerFit thesis

1.5 Contributions of the Thesis

The principal contribution of this thesis is the design and implementation of an integrated, controlled recruitment workflow rather than the invention of a new machine-learning model. The contributions are summarized as follows:

- An integrated recruitment platform that combines a job portal, CV–JD matching, personalized recommendation, application management, and role-specific operational workspaces.
- A transparent text-scoring pipeline using TF-IDF and cosine similarity, with normalized results, labels, reasons, validation signals, and explicit separation between matching and recommendation objectives.
- A feedback-learning workflow based on the Rocchio method, designed to update learned representations from explicit feedback and to support reproducible recomputation rather than uncontrolled cumulative modification.
- A policy-driven AutoFit layer that separates relevance scoring from business actions and applies Human-in-the-Loop controls, user consent, operational constraints, and previous interaction states.
- An actionable email mechanism with expiring, hashed tokens, a non-mutating GET confirmation page, POST redemption, and action-state tracking.

- A traceable evaluation structure that distinguishes algorithm experiments from functional, integration, acceptance, performance, reliability, and security evidence, while explicitly documenting threats to validity.

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Thesis objective	Implemented contribution	Evaluation evidence
Analyze IT recruitment workflows	Role-specific requirements and Human-in-the-Loop boundaries	Use cases and traceability matrix
Implement CV–JD matching	TF-IDF, cosine scoring, explanations, and persisted Matchings	Algorithm benchmark and backend tests
Learn from feedback	Rocchio updates followed by scheduled recomputation	Baseline-versus-learned ranking metrics
Support controlled automation	AutoFit policy, quota, cooldown, notification, and audit controls	Security tests and P0 E2E flows
Deliver an integrated platform	Spring Boot, React/TypeScript, PostgreSQL, and Flyway	Build, runtime health, and browser evidence

Table 1.1. Mapping of objectives, contributions, and evidence

1.6 Thesis Organization

The thesis contains six chapters. Chapter 1 introduces the problem, objectives, scope, and contributions. Chapter 2 presents the theoretical background and related work, including TF-IDF, cosine similarity, Rocchio feedback, ranking metrics, Human-in-the-Loop control, and relevant security foundations.

Chapter 3 covers requirements, use cases, architecture, data design, security, and deployment. Chapter 4 explains the implementation. Chapter 5 presents the evaluation method and verified results, and Chapter 6 discusses achievements, limitations, future work, and the final conclusion.

CHAPTER 2. THEORETICAL BACKGROUND AND RELATED WORK

2.1 Recruitment Information and CV–Job Description Matching

Recruitment matching is an information-retrieval and decision-support problem in which heterogeneous evidence about candidates and vacancies must be compared. A curriculum vitae commonly describes education, employment history, projects, technical skills, certifications, languages, and contact information. A job description describes responsibilities, required and preferred skills, seniority, location, employment conditions, and organizational context. Some attributes are structured, such as years of experience or location, while others appear in free text and may use inconsistent terminology. Consequently, an exact database filter is useful for hard constraints but insufficient for ranking the overall relevance of a candidate and a position.

The same information supports different user tasks. From the candidate perspective, the system retrieves and ranks jobs that fit a CV or desired profile. From the recruiter perspective, it ranks applicants or discovers potential candidates for a specific vacancy. These directions share document representations and similarity functions, but they are not identical business decisions. A relevant job recommendation does not imply that an application has been submitted, and a high candidate–job similarity does not imply that a recruiter should accept the candidate. The score is therefore evidence for prioritization rather than a final employment decision.

Recruitment data also has domain-specific ambiguity. A technology may be expressed by an abbreviation, a product name, or a broader skill family; job titles vary across companies; and the same term may have different importance at junior and senior levels. Open recruitment corpora are relatively uncommon because CVs contain personal information. The Djinni Recruitment Dataset illustrates both the scale of the matching problem and the value of anonymized, documented corpora, providing more than 150,000 jobs and 230,000 candidate profiles in English and Ukrainian [6]. CareerFit is narrower: it focuses on IT recruitment and uses explicit fields and normalized text so that matching behavior can be inspected within the project scope.